KSENIA SHPERLING

SOFTWARE ENGINEERING STUDENT | APPLIED AI & PYTHON | CLASS OF 2027

Moscow, Russia | shperling05@inbox.ru | +7 987 397 11 14
Available September 2027 | Willing to relocate to Dubai and work in a hybrid arrangement
Portfolio: kssperling.github.io/resume | GitHub: github.com/kssperling

PROFILE

Software Engineering student graduating in 2027 with practical experience building AI-enabled application features. Implemented embedding-based semantic search and React interfaces for a team recruitment platform, and contributed to Python/FastAPI backend and LLM integrations. Junior Analyst supporting an AI assistant through testing, debugging and data preparation. Mathematical analysis teaching assistant with a foundation in probability, statistics and linear algebra.

TECHNICAL SKILLS

Programming: Python, C#, SQL, C++; JavaScript/TypeScript through React project work.

AI & data: LLM API integration, OpenAI-compatible APIs (BotHub), embeddings, semantic search, PostgreSQL/pgvector, scikit-learn, Pandas, NumPy, data preprocessing and model evaluation.

Application development: FastAPI, REST APIs, JSON, React, PostgreSQL, MySQL, Docker, Git; software testing, code review, debugging, algorithms and data structures.

MACHINE LEARNING COURSEWORK

Completed regression, classification, text classification and clustering assignments using Python, NumPy, Pandas and scikit-learn. Implemented coursework versions of logistic regression, decision trees, gradient boosting and K-means. Used TF-IDF, cross-validation, GridSearchCV and Optuna; evaluated models with F1, ROC-AUC, precision/recall and regression errors. Explored PCA, t-SNE, DBSCAN and permutation feature importance. Wrote unit tests for a custom scoring function.

EXPERIENCE

Junior Analyst | Rikor Holding

Oct 2023 - Present

- Contribute to Lara, an AI assistant, through software testing, code review, debugging, and bug fixing.
- Clean and preprocess data for analysis; translate tasks into algorithmic steps and use AI tools to help draft tests and test examples.

Teaching Assistant, Mathematical Analysis | HSE University

Sep 2024 - Present

- Support 30+ first-year students through consultations, homework grading, and written and oral assessments.
- Explain mathematical concepts and problem-solving approaches clearly; respond to student questions.

SELECTED PROJECTS

AI-Powered Job Platform | University Team Project

Python, FastAPI, React, TypeScript, PostgreSQL, pgvector

- Developed the frontend for a recruitment platform with AI-assisted resume analysis, job matching, and interview preparation features.
- Implemented embedding-based semantic search for jobs and resumes using PostgreSQL and pgvector.
- Contributed to backend development and LLM API integration through BotHub using the OpenAI SDK.

Visual Novel Framework | Independent University Project

C#, Unity, Live2D

- Independently wrote the application code, including a dialogue script parser, extensible command system, and character management.
- Implemented animated text display, coroutine-based dialogue and command execution, character transitions, and Live2D expression and animation controls.

EDUCATION & LANGUAGES

HSE University, Faculty of Computer Science | Moscow, Russia
Bachelor's degree in Software Engineering | Expected graduation: 2027

Credit-weighted average: **7.67/10** (self-calculated from graded entries in the available transcript).

Selected grades: Mathematical Methods for Data Analysis: 9/10; Advanced Data Analysis, AI and Generative Models examination: 8/10. Foundation in probability, statistics and linear algebra.

Languages: Russian - Native | English - Upper-intermediate (B2)

HONORS

Prize Winner, Final Round - RTU MIREA Multidisciplinary Olympiad for school students, 2023.